



Meniscal Transplantation

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INTRODUCTION

Our understanding of the role of the meniscus has evolved over many years. Dating back to 1948, Fairbank et al. described changes to the knee joint such as narrowing of the joint space, squaring of the femoral condyle, and ridge formation following meniscectomy.¹ This led to his conclusion that a simple meniscectomy is not “wholly innocuous” and it leads to changes in joint mechanics that can be detrimental.

Through the years, extensive research has improved the understanding of meniscal anatomy, biology, mechanics, function, and pathology. This knowledge has adapted our treatment approaches. Prior to Fairbank’s work, total meniscectomies were thought to be the ideal choice for knee pain due to a meniscal injury.

Starting in the 1980s, multiple publications identified the benefits of a partial meniscectomy over a total meniscectomy. These benefits included lower contact pressures, less long-term degenerative changes on radiographs, and improved patient satisfaction.^{2–5} Since this research, trends toward meniscal preservation developed including partial meniscectomy or meniscal repair.⁶

Even with all the advances made over the years, there are still complex situations when meniscal preservation is not possible. For these difficult scenarios, meniscal transplantation may be an option. Dating back to 1989 when the first series of meniscal transplantations was published, there has been an increasing acceptance of meniscal allografts used for situations where the meniscus is nonfunctional, symptomatic, and irreparable.⁷ This chapter offers a comprehensive review of meniscal transplantation including preoperative workup, indications, techniques, and outcomes.

HISTORY

A thorough history is imperative for accurate diagnosis and reasonable treatment recommendations. A detailed understanding of a patient’s pain including duration, quality, location, description, and alleviating and aggravating factors can give a valuable insight into their ailment. Other important factors include locking, catching, or swelling that indicates there is a mechanical issue with the knee. Reports of instability with increased joint line tenderness localized to the involved compartment are sometimes seen with meniscus deficiency. Intermittent swelling may also be present, specifically with an increased level of activity. Emphasis

should be placed on reports of increasing pain in a particular compartment, because this pain may be associated with increased compartment loading and risk for progressive chondral damage.

Respecting patients’ goals of therapy is also important and can help guide the aggressiveness of treatment. Their age and activity level should be taken into consideration as this can alter the optimal treatment strategies for patients. A thorough surgical history regarding the knee can make future surgical procedures more difficult but also allow the surgeon to understand what state the knee is currently in. It is helpful to obtain prior operative reports as well as arthroscopy images if available to gain a better understanding of the status of the knee.

PHYSICAL EXAMINATION

The physical exam is key to determining the location of patients’ symptoms. Correlating symptoms with the location of the affected compartment is critical to a successful result.

Additionally, assessment of the patient with meniscal deficiency and/or articular cartilage damage must occur in the context of the alignment, stability, and meniscal status (Fig. 95.1).⁸ The physical exam and imaging should focus on understanding the entire context of these factors.

Inspecting the standing alignment of the leg and assessing for varus or valgus angulation is essential to understanding the biomechanical context in which a patient’s meniscal deficiency exists and should be substantiated with appropriate imaging. Assessment of range of motion (ROM) is helpful in differentiating the arthritic knee from the healthy knee. Significant motion loss is a contraindication to joint-preserving procedures. The presence or absence of an effusion is also a critical “vital sign” for how the knee is behaving. Joint effusions can be due to many etiologies including trauma, systemic diseases, infection, and mechanical irritation. A knee that is actively delaminating cartilage often presents with recurrent effusions and therefore large effusions can be indicative of progressive articular cartilage loss in this setting.

Focal tenderness along the joint line is also helpful in correlating symptoms in the affected compartment.⁹ Although there are a variety of provocative physical exam maneuvers described for detecting meniscal pathology, they are better suited to detecting meniscal tears rather than meniscal deficiency.¹⁰ A comprehensive ligamentous exam is essential, as untreated instability is a contraindication to meniscal reconstruction.¹¹

Abstract

Meniscal deficiency in young active patients can lead to disastrous outcomes, including early onset arthritis. These patients pose a difficult problem for orthopaedic surgeons as they would otherwise be too young and active for joint replacement options. Meniscal allograft transplantation is a viable option for these patients. One must use the correct indications in deciding if these patients are candidates for a meniscus transplant. Several factors should be taken into consideration, including meniscal status, alignment, and ligamentous stability. Concomitant procedures to realign or stabilize the knee may be indicated in certain circumstances. This is a comprehensive review of meniscal allograft transplantation including indications, preoperative workup, and the author's preferred technique.

Keywords

meniscus
meniscal allograft transplantation
meniscal deficiency
high tibial osteotomy
distal femoral osteotomy

Lastly, since the vast majority of these patients have had prior surgery, assessment of prior incisions on the knee is key to consider as this may limit or alter the surgical plan.

IMAGING

The history and physical exam should guide further workup of the patient, which in most cases entails additional imaging.

Radiographs

A complete radiographic evaluation should include standing anterior–posterior and 45-degree flexed posterior–anterior views as well as lateral and skyline views of the affected knee. Full-length standing x-rays of both legs is important to assess the overall alignment of the limb and evaluate for genu varum or valgum.

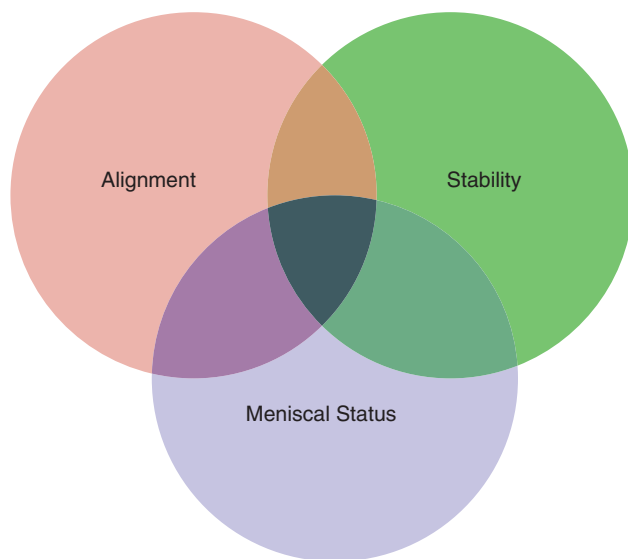


Fig. 95.1 Venn diagram of the key components of the physical exam.

Magnetic Resonance Imaging

Magnetic resonance imaging (MRI) is a valuable tool in the workup of these patients. It allows the physician to evaluate the soft tissue structures of the knee including the meniscus, cartilage, and ligaments, which are all important to consider when formulating a treatment plan. The MRI can also help assess the integrity and amount of residual meniscus (Fig. 95.2).

Bone Scan

Bone scans are sometimes helpful and often underutilized. In view of other normal studies such as an MRI, an increased uptake on a bone scan may suggest compartment overload and impending chondral damage, thus increasing the importance of attempting to restore the compartment load-sharing through meniscal allograft transplantation (MAT), realignment osteotomy if indicated, or both.

DECISION-MAKING PRINCIPLES/INDICATIONS

Decision-making in joint preservation must consider patient factors, the pathophysiology, and any other contributing factors. The knee should be viewed holistically as an organ. Understanding the mechanical alignment, stability, and articular cartilage status is key for a successful outcome.

Patient

Ideally, meniscal transplantation is best reserved for younger patients with compartment-specific knee pain who are meniscal deficient from prior near-total or total meniscectomy, but who still have preserved articular cartilage. It is ideally suited for patients who would like to remain active and have minimal to no arthritic changes.¹²

Pathophysiology

The joint should be evaluated holistically when considering a patient for a meniscal transplant. There is conflicting evidence

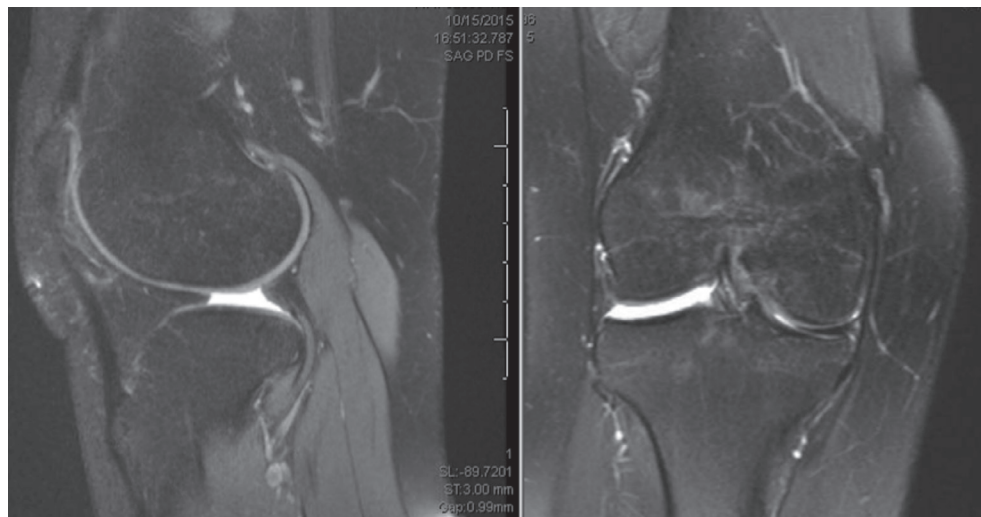


Fig. 95.2 T2-weighted magnetic resonance imaging of an 18-year-old female with right knee lateral meniscal deficiency.

regarding how arthritic changes may affect the outcome of a meniscal transplant. Some believe that arthritic degeneration precludes patients from meniscal transplantation; however, other studies have shown pain relief in patients even with Outerbridge grade III and IV lesions as long as they are isolated.^{8,13,14} These higher-degree lesions can and should be addressed at the time of surgery through various methods such as microfracture, osteochondral allografts, articular cartilage paste grafting, or autologous chondrocyte implantation.^{8,14,15} Bony changes such as flattening or osteophyte formation commonly seen with advancing osteoarthritis may alter the shape of the condyle and lead to poor outcomes with meniscal transplantation.¹⁶ Instability of the knee should also be taken into account for meniscal deficient patients and should be addressed either in a staged or concomitant fashion. Anterior cruciate ligament (ACL) reconstruction at the time of meniscal transplant is an acceptable treatment approach and shows similar outcomes to patients who underwent an isolated meniscal transplant.¹¹ In addition to its load-sharing and articular cartilage preservation role, the stabilizing effect of the medial meniscus should not be overlooked. The posterior horn of the medial meniscus is an important contributor of secondary stabilization to anterior translation of the tibia relative to the femur in the ACL deficient and ACL reconstructed states.¹⁷ This secondary stabilization is an important factor to consider when assessing the ACL deficient patient, particularly in the revision setting.¹⁸

Contributing Factors

Limb alignment is one of the most important issues to evaluate when considering meniscal transplantation. As discussed previously, full-length x-ray can highlight joint deformities such as genu varum or valgum, which increases joint pressures through the respective compartment of the knee.¹⁹ Realignment procedures such as high tibial or distal femoral osteotomies can be performed concomitantly or prior to meniscal transplantation to decrease contact pressures through the compartment (Fig. 95.3).^{16,20} Although patients who undergo concomitant procedures have favorable outcomes, it is still unclear if these procedures should ideally be performed concomitantly or in a staged fashion if osteotomy alone is insufficient to relieve symptoms. If done in a staged fashion, the osteotomy should be performed first and consideration of meniscal transplantation should be reserved for the persistently symptomatic patient.

Contraindications

While meniscal transplantation can be beneficial when applied to the above indications, there are several contraindications including prior or current joint infection, advanced osteoarthritis, obesity, synovial disease, or rheumatoid arthritis.¹²

TREATMENT OPTIONS

A young patient with symptomatic meniscal insufficiency is a challenging clinical scenario. Options vary and considerations must take into account multiple factors such as patient goals, prior surgeries, and status of the residual meniscus. Most of



Fig. 95.3 Anterior-posterior radiograph status after a distal femoral osteotomy to correct genu valgum effectively unloading the lateral compartment.

these patients have had prior sub-total meniscectomies and are therefore not candidates for a meniscal repair or further débridement.

Nonoperative Management

Conservative management should always be attempted first in patients with symptomatic meniscal deficiency. A course of nonsteroidal antiinflammatories, rest, ice, and elevation can help calm down acute inflammation and reduce pain. Some may benefit from strengthening, stretching, or formal physical therapy.²¹ Unloader bracing is an effective way to reduce pressures unilaterally across the knee joint and enhance stability; however, compliance among patients is universally poor.^{22,23} Steroid injections in younger meniscal deficient patients should be used sparingly as recent studies have shown that use may be deleterious to hyaline cartilage.²⁴

Operative Management

Young patients with symptomatic meniscal deficiency have limited options. These patients have already undergone a complete or near complete meniscectomy, so further débridement or repair is not an option. Operative considerations should look to address a combination of factors in these patients. Instability and malalignment must be corrected either before or concomitant to the meniscal transplant as these factors place the transplant at risk of failure.^{11,16,20} In patients who have malalignment, often osteotomy alone is sufficient to decrease loads across the compartment and meniscal transplantation is not necessary. Unicompartmental or total knee arthroplasty is an alternative option for meniscal deficient patients; however, this may limit younger patients' activities and likely subject them to a revision surgery in their lifetime.²⁵



Authors' Preferred Technique

Proper allograft sizing is critical to ensure that the meniscus can distribute and share the load across the cartilage. The patient's radiographs are compared with the donor radiographs to choose an appropriate size length and width meniscus. The meniscus allograft is fresh frozen; compared to cryopreservation, the freezing process kills the graft cells and reduces the possibility for an immune response.

Positioning and Diagnostic Arthroscopy

The patient is positioned supine on the operative table. An exam under anesthesia is performed to confirm full knee passive ROM and a thorough ligamentous examination of both knees is performed to confirm knee stability.

The operative knee is positioned with a lateral thigh post, as well as a foot post to hold the knee in 90 degrees of flexion. A tourniquet is placed, but only inflated if necessary. Standard anterolateral and anteromedial portals are created and a diagnostic arthroscopy is performed to diagnose any associated injuries, confirm knee stability, and rule out any advanced arthritic disease.

If the patient is a candidate for meniscus transplant given the above stated factors, the procedure is continued on and the graft is open and thawed.

Lateral

For lateral meniscal transplantation, the author prefers an all soft tissue technique. The graft is prepared, rinsed, and threaded with three sutures: one in each root and one at the posterior horn–body junction. The graft and all sutures are thoroughly irrigated using antibiotic impregnated normal saline. Intra-articular preparation is performed by removing the remaining meniscus down to a thin rim at the capsule (Fig. 95.4). Ideally a rim of native meniscus should be preserved to minimize the risk of meniscal allograft extrusion. Confirmation of circumferential capsular bleeding is obtained.

An approximate 3-cm incision is made parallel to and immediately posterior to the lateral collateral ligament (LCL), one-third

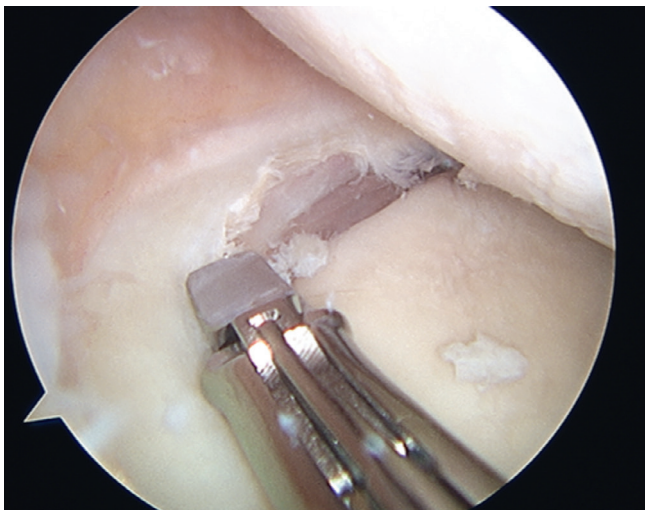


Fig. 95.4 Preparation of remaining meniscus.

above and two-thirds below the joint line. The internervous plane between the biceps femoris and iliotibial band is identified. The lateral head of the gastrocnemius is identified and elevated off of the posterolateral capsule. A Henning retractor is then placed in the posterolateral aspect of his knee.

The posterior root for the lateral meniscus is then drilled. A low-profile multiuse marking (Arthrex) hook is placed through the medial portal. An approximate 3-cm incision is made over the anteromedial tibia. Under direct visualization, a 6-mm Flip-Cutter (Arthrex) is advanced. Once in the anatomic position of the root, a curved curette is placed over it to protect the ACL and the posterior neurovascular structures. A 5-mm blind socket is created in that location to accommodate the posterior root. A passing stitch is placed through that tunnel. In a similar fashion, an additional passing stitch is placed using an inside-out technique just lateral to the popliteus. This correlates to two of the three sutures previously placed in the graft: one in the posterior root and one at the popliteal hiatus.

A 5-cm arthrotomy is made in line with the lateral portal. With the scope in the medial portal, the passing stitches from the root and popliteal hiatus are passed through the previously placed passing stitches in the transplant and the meniscal allograft is reduced. The root is held in place using tension, and numerous vertical mattress sutures are used with zone-specific cannulas in an inside-out fashion. These are alternated between the superior and inferior aspects of the meniscus and spaced approximately 3 to 5 mm apart. The allograft is tied to any remaining remnant of the native meniscus.

Once all sutures are passed, the anterior root footprint is identified and drilled directly through the arthrotomy. A small Beath pin is advanced into the anterior root insertion and then over-reamed with a 4.5-mm drill to create a small bleeding surface. The sutures from the anterior root are passed through the anteromedial tibia with a free needle. Suturing of any remaining anterior portion of the graft is completed using a freehand #2 FiberWire (Arthrex) suture through the arthrotomy.

All of the peripheral meniscus sutures are tied down. The knee is cycled through a full ROM to confirm meniscus stability as well as absence of catching or irregularity of the meniscus. A small area in the anteromedial tibia is cleared and a 4.5-mm screw with a washer (Synthes) is placed as a post and the two meniscal root sutures are secured down over the post.

Arthroscopic visualization is used to confirm coverage of the lateral plateau and stable roots. The entire allograft is probed. The knee is once again cycled through full ROM to confirm stability.

Medial

For medial meniscal transplantation, the author prefers a two bone-plug technique. The graft is prepared with two separate bone plugs of the desired diameter and length (the author prefers 6 mm in diameter and 5 mm in depth). Two sutures are prepassed through the allograft to aide in meniscal delivery and reduction—one through the posterior root and the other through the meniscus at the posterior horn–body junction.

Once again, the compartment is first prepared by removing the remaining meniscus, leaving a thin rim at the capsule and

confirming capsular bleeding. The knee is brought into 90 degrees of flexion and a notchplasty is performed. The medial tibial spine is débrided, if necessary, for access to the posterior medial meniscus root. This is performed below the posterior cruciate ligament (PCL) taking great care to protect the PCL fibers at all times.

An ACL guide or multiuse marking hook (Arthrex) is placed in the center of the posterior root of the medial meniscus. A small anteromedial incision is made and the guide pin is advanced through the anteromedial aspect of the tibia up into the footprint of the posterior horn of the medial meniscus. A 6-mm FlipCutter (Arthrex) is advanced and a 5-mm blind socket is created. A passing stitch is placed through that tunnel.

An inside-out repair technique is preferred for suturing the meniscus to the capsule and meniscal rim. A standard exposure for an inside-out meniscal repair is performed. A Henning retractor is placed in the posteromedial aspect of the wound for suture retrieval.

A Hewson suture passer is passed through the bone tunnel in the posterior root. The medial portal incision is extended and a small medial arthrotomy is performed. The suture passer is retrieved through the medial arthrotomy and used to shuttle a passing stitch through a posterior root tunnel to aid in reducing the meniscus.

With the scope in the medial incision, a zone-specific cannula is used at the junction of the posterior horn and midbody segment and a Nitinol wire with a free loop is passed and retrieved through the posteromedial incision. This will serve to shuttle the prepassed suture at the posterior horn–body junction in the transplant.

The sutures for the posterior root are shuttled using the passing suture in the posterior root tunnel. The stitch placed at the junction of the posterior horn and midbody segment is then passed through the Nitinol wire with the loop and shuttled through the capsule.

The knee is then brought into flexion and valgus. Gentle traction is applied to the stitches and using a blunt trocar, the meniscus is reduced. Anatomic placement is confirmed.

Using zone-specific cannulas, sutures are placed at the periphery of the graft. The posterior horn and body sutures are performed using an inside-out approach, while the anterior sutures are performed using an outside-in approach or directly through the arthrotomy. These are all placed in vertical mattress fashion.

The anterior root is similarly drilled through the arthrotomy. A Beath pin is drilled in the anterior root site and reamed to a depth of about 15 mm. Using a free needle, the two sutures placed in the anterior root are passed directly through the bone. The bone plug is manually reduced and tapped down so that it is flush with the articular surface. The root sutures can then be tied down over a bone bridge anteriorly or to a button or post.

The knee is cycled through a full ROM to confirm meniscus stability as well as absence of catching or irregularity of the meniscus (Fig. 95.5). Arthroscopic visualization, including a Gilquist portal, is used to confirm coverage of the medial plateau and stable roots. The entire allograft is probed. The knee is once again cycled through full ROM to confirm stability.

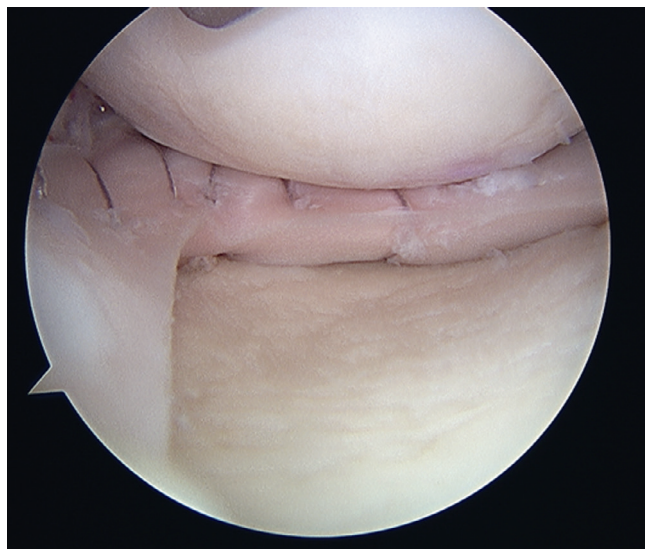


Fig. 95.5 Well-reduced meniscal allograft with sutures in place.

Single Bone Plug and Trough

MAT on the lateral side can also be performed with a single rectangular bone block and trough as described by Goble et al.²⁶

The lateral meniscus allograft is prepared with a single rectangular bone block connecting the anterior and posterior horns. Various guide sutures are preinserted into the allograft. Goble et al. describes the critical suture placement as the posterolateral corner, the edge of the anterior horn (15 mm lateral to its insertion), and two sutures in the common bone block (10 cm apart). The bone block is prepared until it matches a chosen sizing block (8, 9, or 10 mm). A silicone template is then trimmed to match the allograft bone block. The silicone template is inserted, and the trough is made using a power burr (beginning 1 cm posterior to the anterior edge of the lateral tibial plateau and progressing posteriorly). Using an ACL tibial guide, two transosseous holes are placed 10 mm apart in the middle of the trough (matching the single bone block sutures) and exiting the tibia anteriorly. The meniscal allograft is inserted and the two sutures in the bone block are pulled through their respective transosseous hole. A spinal needle and suture are inserted outside-in through the posterolateral corner of the knee, and the preinserted suture in the posterolateral edge of the allograft is then pulled through. A combination of pulling on the three guide sutures, gently placing the graft into the trough, and varus stress on the knee will anatomically reduce the allograft. Guide sutures are tied, and any other additional peripheral sutures are placed.²⁶

The technique for a single bone plug for the medial meniscus is similar to the lateral meniscus technique, with a few exceptions. The medial meniscus horns are farther apart compared to those of the lateral meniscus, and therefore the single bone block connecting the two horns is much longer. The medial trough is performed starting at the anterior margin of the tibial plateau below the fat pad, and the bone is débrided in line with the medial tibial spine axis. Lastly, it is frequently necessary to strengthen the single bone block with a 1/16" guide wire through the center of the bone plug.²⁶

POSTOPERATIVE MANAGEMENT

There is significant variability in surgical rehabilitation protocols for postoperative management after MAT. In their systematic review, Rosso et al. found full weight-bearing was allowed at an average of 6.2 weeks, while full ROM was allowed at an average of 6.3 weeks. A total of 43% of studies (24 of 55) reported the use of a brace, while 16% (9 of 55) reported the use of a continuous passive motion (CPM) machine.²⁷

In the literature's largest case series of meniscal allograft transplantation (included in the systematic review above), Kim et al. describes his postoperative protocol. Isometric strengthening exercises begin immediately after surgery. A CPM device is used postoperative day 1, 0 to 60 degrees for the first 3 weeks, up to 120 degrees during weeks 4 to 6, with full flexion allowed during weeks 8 to 12. The patient is non-weight-bearing for 3 weeks, partial-weight-bearing at 4 weeks, and full-weight-bearing at 7 weeks.²⁸

The author's postoperative protocol consists of non-weight-bearing for 6 weeks. The patient is allowed to toe touch when standing still, but not for ambulation. The patient is limited from 0 to 90 degrees for the first 3 weeks and is progressed thereafter. The patient initiates a CPM device starting on postoperative day 1, starting from 0 to 30. The patient is instructed to increase by 5 degrees per day up to 90 degrees for the first 3 weeks. Thereafter, the patient will start progressing until they reach 120 degrees.

RESULTS

MAT outcomes are confounded by differences in surgical fixation, surgeon technique, and concomitant procedures.²⁹

Most literature report an improvement in pain, function, and patient satisfaction. Rosso et al. included 55 papers in his systematic review of MAT, with an average follow-up of 53 months. The weighted average Lysholm score improved from 56 preoperatively to 83 and the weighted average Visual Analogue Scale (VAS) pain score improved from 6.4 to 2.4, although some authors described a worsening of the results over time. Rosso et al. found an 82% average patient satisfaction.²⁷ Elattar et al. included 44 studies (352 grafts) in his systematic review. The average preoperative Lysholm score improved from 44 to 77, the mean Tegner activity score improved from 3 to 5, and the overall VAS went from 48 to 17. These authors also noted the improvement in these scores tended to gradually decline over time. Elattar et al. found an 89% average patient satisfaction.³⁰ Hergan et al. reported on 14 studies in his systematic review. The authors noted improvements in objective and subjective outcome measures in relatively young patients without instability, malalignment, or substantial cartilage damage. The authors concluded MAT may alleviate knee pain, improve knee function, and have good patient satisfaction if done on the optimal candidate.³¹

One of the most important outcomes for MAT is its influence on preventing or delaying the onset of osteoarthritis. It is unclear at the current time if MAT successfully alters the natural history of the meniscus-deficient knee. González-Lucena et al. studied 33 patients with a mean follow-up of 6.5 years. The authors

observed no significant joint space narrowing between the preoperative (mean 3.19 mm) and postoperative periods (mean 3.21 mm).³² In the Hommen et al. study of 20 MAT, 10 of 15 allografts revealed significant radiographic joint space narrowing, and 12 allografts showed progression of degenerative joint disease in the transplanted compartment. Of the seven patients who received MRIs, all grafts had moderate meniscus shrinkage and five had grade III signal intensities.³³ Saltzman et al. reported on six patients evaluated with radiographs at a mean last follow-up of 8.5 years. The authors reported preservation of joint space narrowing in two patients, minor-to-mild joint space narrowing in two patients, and progression of joint space narrowing with mild-to-moderate change in two patients.³⁴ Although some have suggested MAT may prevent the progression of osteoarthritis, this effect has not been definitively demonstrated and therefore no final conclusions can be drawn.

Few studies compare bone plugs and suture-only fixation. In the Abat et al. study of 33 suture-only grafts and 55 performed with bone plug fixation, a higher percentage of extruded meniscal tissue was seen in the suture-only group compared to the bone plug method ($P < .001$). However, there was no difference in functional score between the two groups.³⁵ In the Hommen et al. report of 13 suture-only grafts and 6 fixed with bone plugs, the two methods yielded similar postoperative Lysholm results ($P = .3228$) and postoperative pain scores ($P = .7653$).³³ The literature is scarce in comparing suture-only techniques and bone plug fixation, with the existing data reporting no difference in functional outcomes or pain scores between the two, although the suture-only technique has a higher rate of extrusion.

COMPLICATIONS

The definitions of failure vary within the literature, likely resulting in differing survival times and survival rates. Noyes et al. reported on 72 consecutive medial and lateral MATs, defining failure as reoperations related to the transplant, signs of meniscal tear on clinical examination, loss of joint space in the involved tibiofemoral compartment on 45 degrees weight-bearing posteroanterior (PA) radiographs, or grade 3 signal intensity or extrusion greater than 50% of the meniscus transplant width on MRI. The estimated probability of survival was 85% at 2 years, 77% at 5 years, 45% at 10 years, and 19% at 15 years.³⁶ In the Verdonk et al. study of 100 consecutive MATs, failure was defined as moderate or severe occasional or persistent pain or as poor knee function. The mean cumulative survival time (11.6 years) was identical for the medial and lateral allografts. The cumulative survival rate for medial transplants was 86.2% at 5 years, 74.2% at 10 years, and 52.8% at 14.5 years. The cumulative survival rates for lateral allografts was 90.2% at 5 years, 69.8% at 10 years, and 69.8% at 14.5 years. These did not differ significantly ($P = .733$).¹³ van Arkel et al. reported on the outcomes of 63 consecutive MATs, defining allograft failure as persistent pain, an unsuccessful knee assessment scoring system (KASS) result, a poor Lysholm score, or a detached allograft. The mean survival time for the lateral allografts was 111 months versus 69 months in the medial allografts. The cumulative survival rate

of the lateral allografts was 76% versus 50% in the medial allografts ($P = .004$).³⁷

In their systematic review of 55 studies, Rosso et al. found 176 complications in 1666 grafts, with a complication rate of 10.6%. The most common complication was an allograft tear (105 cases, 60% of all complications). Synovitis or effusion was reported in 54 cases (31% of all complications), superficial infections were reported in 11 cases (6% of all complications), a decrease in ROM was reported in five cases (3% of all complications), and deep infection was reported once.²⁷

CONCLUSION/FUTURE CONSIDERATIONS

Although technically challenging, meniscal allograft transplantation is a viable option in a young patient with a symptomatic, meniscal-deficient knee. Appropriate patient selection is key to a successful outcome. The patient's symptoms should be localized to the affected compartment and the knee should be absent of or have minimal arthritic changes. Knee stability should be

assessed and accounted for if it is compromised. Malalignment should be evaluated and corrected in a staged or concomitant fashion if indicated, as uncorrected malalignment results in increased forces through the respective compartment and a sub-optimal environment for allograft survival.

Although MAT's ability to improve pain and function has been established with high overall patient satisfaction, there is still much to be determined. With complication rates reported up to 10% and early failure rates as high as 15%, further study is necessary to establish how to prevent these undesirable outcomes. A standardized definition of MAT failure is needed to better establish complication and failure rates and to enable comparison of surgical techniques. Lastly, the ideal outcome of meniscal reconstruction is chondroprotection and the prevention of osteoarthritis. Further refinement of techniques, added study of the role of concomitant procedures, and the role of biologics are all areas of future interest and exploration to achieve this ultimate outcome.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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